Assignment 2: House Price Prediction - Bengaluru Housing Data

- **Objective:** Developed a machine learning model to predict house prices in Bengaluru using features such as area, BHK, number of bathrooms, and location.
- **Data Preparation:**
  - Filtered the dataset to include locations starting with 'P'.
  - Performed data cleaning, including filling missing values for BHK, bath, area, and location.
  - Converted the 'total_sqft' column into a numeric 'area' feature, handling range values appropriately.
  - Engineered a 'price_per_sqft' feature and removed duplicate records.
- **Exploratory Data Analysis (EDA):**
  - Observations indicate a right-skewed distribution of house prices.
  - Strong positive correlations exist between area, BHK, bathrooms, and price.
  - Significant price variation was observed across different locations, highlighting the importance of geography in property valuation.
- **Model Performance:**
  - A Linear Regression model was implemented using One-Hot Encoding for categorical location data.

    - An improved model was developed by removing extreme outliers and applying a log-transformation to the target variable ('price') to handle skewness, resulting in better predictive performance metrics.
- **Conclusion:** The model confirms that physical dimensions (area, BHK) and location are the primary drivers of housing prices in the Bengaluru dataset.